

Multiple Choice (10 marks)

1. If a, b, and c are integers, which of the following conditions is sufficient to guarantee that the expression $a < c \parallel a < b \ \&\& \ ! (a == c)$ evaluates to true?
 - (a) $a < c$
 - (b) $a < b$
 - (c) $a > b$
 - (d) $a == b$
2. Let $l = [1,2,3,4]$. What is $\text{sum}(l)$ in Python 3?
 - (a) 4
 - (b) 1
 - (c) 10
 - (d) 8
3. Let $l = [1,2,3,4]$. What is $\text{max}(l)$ in Python 3?
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 4
4. The boolean expression $a[i] == \text{max} \parallel \!(\text{max} != a[i])$ can be simplified to
 - (a) $a[i] == \text{max}$
 - (b) $a[i] != \text{max}$
 - (c) $a[i] < \text{max} \parallel a[i] > \text{max}$
 - (d) FALSE
5. In Python 3, what is $['a', 'Chemistry', 0, 1][-3]$?
 - (a) a
 - (b) Chemistry
 - (c) 0
 - (d) 1

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, the expression `b[::2]` retrieves the elements at even indices, resulting in `[11, 15, 19]`.
7. True or False: In Python 3, the function `min([1, 2, 3, 4])` returns the smallest number, which is 2.
8. True or False: In Python 3, the `seed([x])` function sets the starting value for generating random numbers.
9. True or False: In Python 3, the expression `'a' + 'ab'` produces an error due to incompatible types.
10. True or False: In Python 3, the output of `print(tuple[0])` for the tuple `('abcd', 786, 2.23, 'john', 70.2)` is `'abcd'`.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the implications of not checking for a zero value of `N` in an algorithm that calculates the average of `N` numbers. When and how will this error be identified during program execution?
12. Consider the list `l = [1, 2, 2, 3, 4]` in Python 3. Explain how using the `set` function on this list affects the output and discuss the significance of sets in programming.

End of Paper